

OEAS 705/805 Advanced Environmental Data Science

Fall 2026

Week 4

Dr. Jessie Turner, jturners@odu.edu

Class times: OCNPS 320, T/Th, 11:00 - 12:15

Office hours: After class T/Th,

And: OCNPS 411, M, 1:00 - 3:00



Schedule

Project outline, Due Monday Oct 5th

→ Post link to your Project GitHub to Canvas

Next week: Machine learning and scki—learn

Homework 3 will be Due Monday Oct 19th

→ Post link to your OEAS805 GitHub to Canvas

Building Packages

You or one of your collaborators might want to make
YOUR OWN PACKAGE

What does that look like?

Building Packages

You or one of your collaborators might want to make
YOUR OWN PACKAGE

- ✓ **Clear Documentation**
Write docstrings and an informative README.
- ✓ **Explicit Dependencies**
One environment with specific package versions and python version documented.
- ✓ **No Absolute Paths**
Use relative project paths like ../filename.ipynb (not paths on your local machine)
- ✓ **Open Access Citation**
Attach an open license and citable DOI.

Sharing Code

Once you have created a new package, how to share?

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Version Control

Git repository
with clean
commit history.



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Open License

Include standard
license (e.g., MIT
or Apache 2.0).



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Get a DOI

Archive releases
on Zenodo or
figshare.



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(Optional) Separate Data

Host raw data in
dedicated
repositories.

Examples

1. gliderpy

U.S. Integrated Ocean Observing System (IOOS) - oceanography

2. coastpy

TU Delft (Netherlands) – shoreline change

3. RivGraph

UT Austin – river networks

4. si_openaccess

Smithsonian – museum data

Examples (In Groups)

Things to evaluate:

- Citable?
- License?
- Clear documentation?
- Strengths
- Weaknesses

